



EARLY WARNING SYSTEM FOR ACADEMIC DEPRESSION USING DATA SCIENCE & ANALYTICS





Early Warning System for Academic Depression Using Data

Science & Analytics

Objective:

The objective of this project is to develop a **data-driven early warning system** that predicts whether a student is at risk of developing **academic depression**. This system helps **educators, counsellors, and institutions** identify high-risk students and take **preventive actions** through timely intervention.

Project Overview:

This project involves analysing student data related to **mental health, academic pressure, and lifestyle habits** to predict depression risk. The model uses the **K-Nearest Neighbours (KNN)** algorithm to detect patterns based on similarity among students. The system aims to support **mental health monitoring** using objective, data-driven methods.

Data visualization will also be used to highlight risk factors and prediction outcomes.



Dataset Features:

The dataset includes the following features for each student:

- Age
- Gender
- City
- Profession
- Academic Pressure
- Work Pressure
- CGPA
- Study Satisfaction
- Job Satisfaction
- Sleep Duration
- Dietary Habits
- Degree
- Have you ever had suicidal thoughts ?
- Work/Study Hours
- Financial Stress
- Family History of Mental Illness
- Depression

Scope of Work:

1. Data Collection and Exploration

- Understand the **structure** and **distribution** of data
- Perform **exploratory data analysis (EDA)** to uncover trends and correlations



2. Data Pre-processing

- Handle **missing values**, **categorical encoding**, and **scaling** of features
- Divide the dataset into **training** and **testing** sets

3. Model Development

- Implement the **K-Nearest Neighbours (KNN)** algorithm
- Choose appropriate **value of K** and tune hyper parameters
- Evaluate the model using:
 - **Accuracy**
 - **Confusion Matrix**
 - **Precision, Recall, F1-Score**

4. Data Visualization

- Use **matplotlib** or **seaborn** (or optionally Power BI) to visualize:
 - **Depression risk distribution**
 - **Feature correlations**
 - **Model prediction results**

5. Insights and Recommendations

- Highlight key **lifestyle** and **mental health indicators** contributing to academic depression
- Provide suggestions for **early interventions**, **counselling**, and **policy changes**



Tools and Technologies:

- **Programming Language:** Python
- **Libraries:** pandas, numpy, matplotlib, seaborn, scikit-learn
- **Visualization:** matplotlib, seaborn or Power BI
- **Algorithm:** K-Nearest Neighbours (KNN)
- **Environment:** Jupyter Notebook or Visual Studio Code

Expected Outcomes:

- A trained **KNN model** that predicts depression risk in students
- Insights into **mental health risk factors**
- Visual tools that support understanding of patterns in **student wellness data**
- A structured system for **early alerts** and **preventive support**

Deliverables:

- Cleaned and pre-processed **student dataset**
- Python code implementing the **KNN model**
- Visual outputs (charts, heat maps, dashboards)
- Final report covering **methodology, results, and recommendations**

Evaluation Criteria:

- Completeness of **data analysis** and **feature handling**
- **Accuracy** and **robustness** of the KNN model
- Clarity and quality of **visualizations**
- Depth of **insights and action ability** of recommendations



Additional Instructions:

- Ensure **data privacy** and handle sensitive data ethically
- Document all **model assumptions and limitations**
- Maintain **versioned backups** of datasets and code
- Validate the system with **unseen data** to check generalizability

Note:

Students are encouraged to collaborate, seek guidance when needed, and ensure the ethical handling of data throughout the project.



www.pydun.com



Email Us:
training@pydun.com



Call Us:
+91 93619 99189

PYDUN TECHNOLOGY PRIVATE LIMITED

15/3 Gokhale Road, Tallakulam, Near Lakshmi Sundaram Hall,
Madurai - 625 002, Tamilnadu. India.